

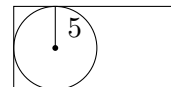
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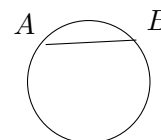
Hutch MATHCOUNTS Handout: Circles 1

1. Chloe Hu walks in a perfect circle, for a total of 10π meters. What is the radius of the circle that she walked? _____
2. Hanning walks in a different perfect circle and notices the area enclosed by her circle is 81π square meters. What is the radius of the circle that she walked? _____

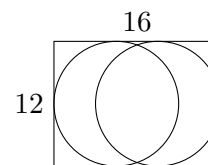
3. Muhammad draws a cool drawing. If the length of the rectangle is twice its width, what is the area of the rectangle? _____



- Armaan sees a weird goat, which we will assume to be perfectly circular
4. and planar. It has a radius 5, and chord AB has length 8. What is the shortest distance from the center of the circle to chord AB ? _____



- In the diagram to the right, the rectangle has length 16 and width 12.
5. Each circle is tangent to three sides of the rectangle. What is the distance between the centers of the circles? _____

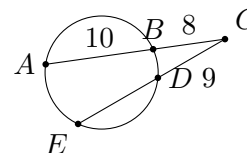


6. After a quick Google Search, Simon learns that the surface area of a sphere is $4\pi r^2$. He takes a perfectly spherical Rubik's Cube and cuts it in half, forming two hemispheres. What is the surface area of each hemisphere? _____

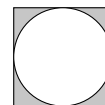
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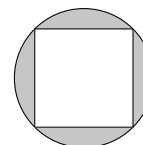
- Aiden, after laughing violently at a 6-7 joke, throws up, and his vomit
7. makes the exact diagram to the right. If $AB = 10$, $BC = 8$, $CD = 9$, and C is a point outside the circle, what is DE ? _____



- Aaron He draws the diagram to the right, consisting of a circle inscribed
8. in a square. The square has a side length of 10. What is the area of the shaded region? _____

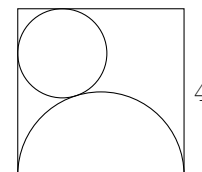


- Aaron Cong tries to copy Aaron He, drawing the diagram to the right,
9. consisting of a square inscribed in a circle. The square has a side length of 10. What is the area of the shaded region? _____



10. Navya thinks of the following cool problem: A right triangle has sides with lengths 8 cm, 15 cm, and 17 cm. A circle is inscribed in the triangle. In centimeters, what is the radius of the circle? _____

- Vincent places a semicircle and a circle inside a square with sides of
11. length 4 cm, as shown. The circle is tangent to two adjacent sides of the square and to the semicircle. The diameter of the semicircle is a side of the square. In centimeters, what is the radius of the circle? _____



12. Ainsley has an epiphany and creates the following brilliant problem: Two circles with radii 16 and 9 are tangent to each other, and are tangent to line at distinct points P and Q . Find the length of PQ . _____